

Teacher Guide — senior / module-10-herbs-aushadhi

IKS Curriculum

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Overview

Herbs — Auṣadhi

Band: Senior (Grades 9–12) · **Sessions:** 10 × 45 min · **Sanskrit:** Auṣadhi

Overview

Six widely-available herbs — Tulsī, Neem, Turmeric, Ginger, Ajwain, Mint — studied through botanical identification, traditional uses from Āyurveda, modern phytochemistry (active compounds), and safety. The module emphasises that kitchen amounts are safe, but therapeutic doses require a qualified practitioner.

NEP 2020 alignment

- §4.6 — Integration of Indian Knowledge Systems
- §4.23 — Experiential learning (plant identification)
- §4.27 — Local context (locally-available plants)

Key terms (this module)

- *auṣadhi* — medicinal plant / herb
- *tulasī* — holy basil — *Ocimum tenuiflorum*
- *nimba* — neem — *Azadirachta indica*
- *haridrā* — turmeric — *Curcuma longa*
- *ādraka* — ginger — *Zingiber officinale*
- *yavānī* — ajwain — *Trachyspermum ammi*
- *pudīnā* — mint — *Mentha* spp.

Prerequisites

Module 2 (Panchabhūta) is recommended before this module. Modules 5 (Bindu Paddhati / dot addition) and 7 (×11 Multiplication) are useful prerequisites for Module 9 and 11.

Materials needed (across 10 days)

- Notebook + pencils per student
- Whiteboard / chart paper
- (See [lesson-plan.md](#) for per-day material lists)

10-day map

- **Day 1** — What is Auṣadhi?: Define auṣadhi and recognise the six herbs of this module.
- **Day 2** — Tulsī — Holy Basil: Identify tulasī by leaf and smell; name the active compound (eugenol).
- **Day 3** — Neem — Bitter Healer: Identify nimba; learn about azadirachtin; understand high-dose safety warning.
- **Day 4** — Turmeric — Golden Root: Distinguish fresh root vs powder; identify curcumin as active compound.
- **Day 5** — Ginger — Warming Root: Distinguish fresh vs dried; gingerol vs shogaol.
- **Day 6** — Ajwain — Carom Seed: Recognise seeds and leaves; identify thymol; traditional digestive use.
- **Day 7** — Mint — Cooling Leaf: Recognise several Mentha varieties; identify menthol.
- **Day 8** — Herbs and the Doṣas: Categorise the six herbs by heating/cooling effect (links to Module 4).
- **Day 9** — Build the Class Deck: Each student contributes one identification card to the class library.
- **Day 10** — Final Share and Assessment: Present the class herb deck; take summative quiz.

Files in this pack

- [lesson-plan.md](#) — detailed 10-day arc
- [sources.md](#) — sources cited in this module
- [teacher-notes.md](#) — common misconceptions, differentiation
- [parent-guide.md](#) — what parents should know
- [days/day-NN-*.md](#) — 10 per-day lesson plans
- [quizzes/formative.md](#), [quizzes/summative.md](#)
- [activities/activity-01-*.md](#)
- [assessments/rubric.md](#), [assessments/project-brief.md](#)
- [student-workbook.md](#)
- [writing-prompts.md](#) (or [oral-recall.md](#) for Primary)
- [homework.md](#)
- [slides/deck.md](#)

Voice

Analytical, comparative, primary-source aware; 20–30 words on average. Pitches the concept first, frames the tradition as historical context, and uses IAST for all Sanskrit terms.

10-Day Lesson Plan

Lesson Plan — Herbs — Auṣadhi (Senior)

10-day arc, 45 min per session.

Day 1 — What is Auṣadhi?

Objective: Define auṣadhi and recognise the six herbs of this module.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 2 — Tulsī — Holy Basil

Objective: Identify tulasī by leaf and smell; name the active compound (eugenol).

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 3 — Neem — Bitter Healer

Objective: Identify nimba; learn about azadirachtin; understand high-dose safety warning.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 4 — Turmeric — Golden Root

Objective: Distinguish fresh root vs powder; identify curcumin as active compound.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 5 — Ginger — Warming Root

Objective: Distinguish fresh vs dried; gingerol vs shogaol.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 6 — Ajwain — Carom Seed

Objective: Recognise seeds and leaves; identify thymol; traditional digestive use.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 7 — Mint — Cooling Leaf

Objective: Recognise several Mentha varieties; identify menthol.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 8 — Herbs and the Doṣas

Objective: Categorise the six herbs by heating/cooling effect (links to Module 4).

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 9 — Build the Class Deck

Objective: Each student contributes one identification card to the class library.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 10 — Final Share and Assessment

Objective: Present the class herb deck; take summative quiz.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Pacing notes

- Senior sessions are 45 min. Adjust if your block is different.
- Days 5 and 9 are buffer days — use to reinforce weaker concepts or extend stronger ones.
- The summative project (see [assessments/project-brief.md](#)) runs in parallel from Day 6 onward.

Differentiation

- **For advanced students:** see [teacher-notes.md](#) for extension prompts.
- **For students needing more support:** simplify activities, do more pair-work, allow oral instead of written responses.

Teacher Notes

Teacher Notes — Herbs — Auṣadhi (Senior)

Common student misconceptions

- *“This is religious, not scientific.”* — Frame the tradition as historical observation. Pañcabhūta-like classifications of matter exist in many cultures. Indian astronomy and math are documented, peer-reviewed historical fields.
- *“The Sanskrit terms are just jargon.”* — Insist on the triple format (Sanskrit / IAST / meaning) once, then use the Sanskrit alone. The terms are precise; the discipline of using them properly is itself a learning outcome.
- *“If it’s traditional, it must work.”* — Distinguish empirical claims (testable) from ethical/aesthetic frames (which need separate justification). Encourage students to ask “how would I test this?”.

Differentiation strategies

- **Extension for advanced students:** offer one optional “deeper read” per day — usually a primary source or modern paper.

- **Support for students needing scaffolds:** more pair-work, allow oral responses, provide partial outlines for writing prompts.
- **Multilingual classrooms:** Sanskrit terms are universal; vernacular equivalents (Hindi, Tamil, Telugu, Bengali, Marathi etc.) should be welcomed.

Safety

- Any activity involving food, plants, fire, or outdoor work needs explicit safety briefing.
- See the relevant day file for activity-specific safety notes.

Honest framing

- Where modern science clearly extends or revises traditional claims, say so.
- Where the tradition's framing is best understood as ethical/aesthetic, present it as such (not as competing science).
- Where attribution is debated (e.g., the “Vedic” status of Tirthaji's mathematics), present the debate honestly.

End-of-module reflection (for teachers)

- Which concepts landed well? Which didn't?
- Which students grew most? Which need a different approach next module?
- Add notes to `teacher-notes.md` for the next teacher who uses this pack.

Sources

Sources — Herbs — Auṣadhi (Senior)

Primary and secondary sources

- Caraka Saṁhitā, *Sūtrasthāna* 1 (introduction to auṣadhi framework)
- Atharva Veda 10.97 — Auṣadhi Sūkta (hymn to herbs)
- Ayurvedic Pharmacopoeia of India (Indian Council of Medical Research)
- P. K. Warrier et al., *Indian Medicinal Plants* (1994)
- Modern phytochemistry reviews in *Journal of Ethnopharmacology*

A note on citation discipline

Per the curriculum's style guide, citations must be verifiable. If a verse number, page, or attribution cannot be confirmed, the source is paraphrased without invented attribution. Where direct quotes appear in Senior-band day files, the source is identified in IAST + edition.

For deeper reading

- The shared glossary at `curriculum/_shared/glossary.md` contains the IAST + meaning for every Sanskrit term used.

- Other modules in this curriculum that connect: see the module README’s “Prerequisites” section.

Day 1 — What is Auṣadhi?

Module: Herbs — Auṣadhi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: define auṣadhi and recognise the six herbs of this module.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *auṣadhi* (IAST: auṣadhi; lit. “medicinal plant / herb”)

Primary source (today’s reading)

From *Caraka Saṁhitā* — see `sources.md`.

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Define *auṣadhi*.** Sanskrit for “medicinal plant” — broader than English “herb.” Includes leaves, roots, barks, seeds, resins. Distinguish from *ahara* (food) and *rasāyana* (rejuvenative).
2. **Six herbs we’ll study.** Show samples: Tulsī, Neem, Turmeric, Ginger, Ajwain, Mint. Pass them around for sight + smell.
3. **The traditional framework.** Each herb has *rasa* (taste), *vīrya* (potency — heating/cooling), *vipāka* (post-digestive effect), and *prabhāva* (specific action). This is the basic *dravyaguṇa* (pharmacology) of Āyurveda.
4. **Source.** Caraka Saṁhitā, Sūtrasthāna 1: “*There is nothing in the world that, applied at the right time and place, is not medicine.*” Read aloud.

5. **The critical caveat, repeated three times.** This module teaches IDENTIFICATION and traditional uses. It does NOT teach therapeutic dosing. Therapeutic use requires a qualified practitioner. State this rule prominently in workbooks.

Activity (15 min)

A scaled version of *Blind Smell-Test Identification* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces What is Auṣadhi?.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 2 — Tulsī — Holy Basil

Module: Herbs — Auṣadhi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify tulasī by leaf and smell; name the active compound (eugenol).

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *tulasī* (IAST: tulasī; lit. “holy basil — *Ocimum tenuiflorum*”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Botanical identification.** *Ocimum tenuiflorum* (also *O. sanctum*). Square stem, opposite leaves with serrated edges, distinctive aromatic oil glands. Multiple cultivars: *Rama tulasī* (green), *Kṛṣṇa tulasī* (purple), *Vana tulasī* (wild).
2. **Active compound.** Eugenol (60-70% of essential oil), ursolic acid, rosmarinic acid. Some research on adaptogenic / anti-inflammatory effects (Cohen, 2014 — *Journal of Ayurveda and Integrative Medicine*).
3. **Traditional uses.** Caraka and Suśruta cite *tulasī* for: cough, cold, low-grade fever, indigestion (in tea form), as a bedside plant for air quality.
4. **Source.** Atharva Veda 10.97 — Auśadhi Sūkta — though not naming *tulasī* specifically, the framing of “herbs as healing beings” begins here. Caraka *Sūtrasthāna* 26 lists *tulasī* formally.
5. **Smell-test demo.** Crush a leaf; pass around. Note clove-like aroma (eugenol). Modern aroma-chemistry: same compound (eugenol) gives cloves their scent.

Activity (15 min)

A scaled version of *Blind Smell-Test Identification* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Tulsī — Holy Basil.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 3 — Neem — Bitter Healer

Module: Herbs — Auśadhi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify nimba; learn about azadirachtin; understand high-dose safety warning.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *nimba* (IAST: nimba; lit. “neem — *Azadirachta indica*”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Botanical identification.** *Azadirachta indica*. Tall tree, ~15-20 m. Pinnate compound leaves with serrated leaflets. Distinctive bitter smell when crushed.
2. **Active compound.** Azadirachtin (in seeds, oil). Nimbidin (in leaves, bark). Quercetin (a flavonoid).
3. **Traditional uses.** Caraka and Suśruta cite *nimba* for: skin conditions, fever, infections, oral hygiene (chewing the twig). External use is well-supported by modern studies.
4. **THE SAFETY WARNING.** Neem at high concentrations (especially neem oil) is documented to be hepatotoxic. Caraka itself notes restrictions. Modern pediatric case-reports (e.g. *Indian Pediatrics* journals) document harm in infants given neem oil. NEVER ingest in concentrated form. State this rule THREE TIMES.
5. **Modern use cases.** Neem-based pesticides (azadirachtin is one of the most studied bio-pesticides). Toothpaste with neem extract. Skin creams. Discussion: “*What separates safe external use from risky internal use?*”

Activity (15 min)

A scaled version of *Blind Smell-Test Identification* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Neem — Bitter Healer.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 4 — Turmeric — Golden Root

Module: Herbs — Auśadhi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: distinguish fresh root vs powder; identify curcumin as active compound.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See `lesson-plan.md` for any day-specific materials)

Vocabulary introduced today

- *haridrā* (IAST: *haridrā*; lit. “turmeric — *Curcuma longa*”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Botanical identification.** *Curcuma longa*. A rhizome (underground stem, not a root). Bright orange when cut. Fresh vs dried (turmeric powder).

2. **Active compound.** Curcumin (~2-5% of dried rhizome). Bright yellow polyphenol. Other compounds: turmerones in the essential oil.
3. **Traditional uses.** Caraka cites *haridrā* for: wound healing (topical paste), anti-inflammatory (golden milk), as a colouring + preservative agent in food. Centuries of culinary use.
4. **Modern research.** Curcumin has been studied extensively for anti-inflammatory effects. Bioavailability is poor without piperine (black pepper) or fat — which is why *haldi doodh* (turmeric milk with ghee) is traditional. Modern preparations use liposomal or piperine-enhanced formulations.
5. **Demo / activity.** Crush a small piece of fresh turmeric on paper. Note the staining (curcumin is a pH indicator — colour changes with vinegar/lemon vs baking soda). This is also chemistry, not just culinary.

Activity (15 min)

A scaled version of *Blind Smell-Test Identification* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Turmeric — Golden Root.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 5 — Ginger — Warming Root

Module: Herbs — Auśadhi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: distinguish fresh vs dried; gingerol vs shogaol.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *ādraka* (IAST: ādraka; lit. “ginger — *Zingiber officinale*”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Botanical identification.** *Zingiber officinale*. Rhizome (like turmeric, not a true root). Light tan skin. Fresh (*ādraka*) vs dried (*śuṇṭhī*) — distinct Sanskrit names because the properties differ.
2. **Active compounds.** Fresh ginger: gingerols. Dried ginger: shogaols (gingerols dehydrate during drying). Both are pungent / “hot.”
3. **Traditional vs modern uses.** Caraka cites ginger for: digestive fire, nausea, cold-induced cough. Modern clinical trials support: nausea (motion sickness, chemo-induced — *Cochrane Reviews*), and mild anti-inflammatory effect.
4. **Two preparations, two effects.** “*Why does Āyurveda distinguish fresh from dried?*” Fresh ginger is heating but mild; dried ginger is more concentrated and warming. Modern: same compound family, different ratios.
5. **Activity.** Smell-test: fresh ginger root vs dried ginger powder. Students rate intensity 1-5 and write one sentence on the difference.

Activity (15 min)

A scaled version of *Blind Smell-Test Identification* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Ginger — Warming Root.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 6 — Ajwain — Carom Seed

Module: Herbs — Auṣadhi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: recognise seeds and leaves; identify thymol; traditional digestive use.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *yavānī* (IAST: *yavānī*; lit. “ajwain — *Trachyspermum ammi*”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Botanical identification.** *Trachyspermum ammi*. Small seeds (closer to caraway than parsley). Distinctive sharp aroma. Leaves of the plant resemble parsley.
2. **Active compound.** Thymol (35-60% of essential oil). Same compound as in thyme. Strong antimicrobial and digestive-stimulating actions.
3. **Traditional uses.** Caraka cites *yavānī* primarily for digestive complaints: indigestion, gas, colic. The traditional remedy: a few seeds chewed after a heavy meal.

4. **Modern research.** Thymol’s antimicrobial effect is well-documented in food science (it’s used as a natural preservative). Limited clinical research on traditional digestive claims, but the underlying chemistry is plausible.
5. **Discussion.** “*Why is the same compound (thymol) found in ajwain and thyme — plants from very different families?*” This is *convergent evolution* in plant chemistry: distantly related plants evolving the same defensive compound. A great cross-cultural pharmacology insight.

Activity (15 min)

A scaled version of *Blind Smell-Test Identification* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Ajwain — Carom Seed.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 7 — Mint — Cooling Leaf

Module: Herbs — Auṣadhi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: recognise several mentha varieties; identify menthol.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *pudīnā* (IAST: pudīnā; lit. “mint — *Mentha* spp.”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Botanical identification.** *Mentha* genus — many species: *M. arvensis* (corn mint, *pudīnā*), *M. spicata* (spearmint), *M. piperita* (peppermint). Distinguish by leaf shape, smell.
2. **Active compound.** Menthol (peppermint), carvone (spearmint). Both terpenoids, both “cooling” perceptually (menthol activates cold-sensing TRPM8 receptors in the mouth).
3. **Traditional uses.** Less prominent in classical Caraka/Suśruta (mint was a later introduction to Indian materia medica). Modern Āyurveda uses for: cooling drinks, digestive aid, breath freshener.
4. **Modern research.** Peppermint oil — well-studied for IBS symptom relief (*BMJ*, multiple trials). Cooling sensation is a TRPM8 channel effect, NOT actual cooling of tissue.
5. **Activity.** Cooling-sensation test: students suck on a peppermint and report on tongue temperature. “*Is the mint actually cool? Or does it just feel cool?*” (It’s the TRPM8 receptor activation — the same nerve that signals real cold.)

Activity (15 min)

A scaled version of *Blind Smell-Test Identification* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Mint — Cooling Leaf.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 8 — Herbs and the Doṣas

Module: Herbs — Auśadhi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: categorise the six herbs by heating/cooling effect (links to module 4).

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *auśadhi* (IAST: auśadhi; lit. “medicinal plant / herb”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Recall doṣa framework (Module 4).** Vāta = motion, Pitta = transformation, Kapha = structure.
2. **Herb classification by doṣa effect.** From Caraka *Sūtrasthāna* 26:
 - **Heating / Pitta-aggravating** (use with care if pitta-dominant): ginger, ajwain, hot pepper.
 - **Cooling / Pitta-pacifying** (good for hot conditions): mint, fennel.
 - **Drying / Kapha-pacifying** (good for congestion): turmeric, neem.
 - **Lubricating / Vāta-pacifying** (good for dryness): ghee-based preparations.
3. **Maps onto modern thermoregulation.** Heating herbs (ginger, ajwain) often increase metabolic rate slightly. Cooling herbs (mint via TRPM8) provide perceptual cooling. The mapping is empirically observed, not purely conceptual.

4. **Worked example.** “If someone has a *vāta* imbalance — dry skin, restless sleep — which of our six herbs would *Āyurveda* recommend?” (Avoid the very dry/cold ones. Ginger in moderate amounts; ghee-based ginger tea.)
5. **Safety reminder.** This is wellness lens, NOT diagnosis. Don’t prescribe based on *doṣa* lens; consult a qualified practitioner.

Activity (15 min)

A scaled version of *Blind Smell-Test Identification* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Herbs and the *Doṣas*.

Wrap-up (5 min)

- One-sentence exit ticket: “What’s one thing you learned today?”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 9 — Build the Class Deck

Module: Herbs — *Auśadhi* · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: each student contributes one identification card to the class library.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *tulasī* (IAST: *tulasī*; lit. “holy basil — *Ocimum tenuiflorum*”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Card-deck project assembly.** Each student claims one of the six herbs as their primary contribution. They produce one detailed identification card:
 - Sanskrit + botanical name
 - Leaf/seed/root drawing (large, accurate)
 - Smell description (concrete words)
 - Three traditional uses with cited sources
 - One active compound and its action
 - One safety note
2. **Tools and references.** Use Caraka Saṁhitā Sūtrasthāna 26 (compiled excerpts in *sources.md*), modern phytochemistry summaries from ICMR-NIN or *Journal of Ethnopharmacology* reviews.
3. **Work-time.** 30 minutes individual work. Teacher circulates to check sources.
4. **Pair-review.** Pairs swap cards; check for accuracy and one quality-improvement suggestion.
5. **Class deck assembly.** Cards laminated (optional) and bound into a class library. Teachers may keep one set for future use.

Activity (15 min)

A scaled version of *Blind Smell-Test Identification* (see *activities/*). Today’s slice: focus on the part of the activity that reinforces Build the Class Deck.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See *homework.md* for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 10 — Final Share and Assessment

Module: Herbs — Auśadhi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: present the class herb deck; take summative quiz.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *nimba* (IAST: nimba; lit. “neem — *Azadirachta indica*”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Class herb-deck presentation (20 min).** Each student shows their card and gives a 1-minute talk on their herb.
2. **Q&A (10 min).** “*Which herb would you most want in your home garden? Why?*” “*Which one surprised you?*”
3. **Summative quiz (20 min).** Vocabulary, identification, doṣa-pairing, safety. See quizzes/summative.md.
4. **Critical-thinking close.** “*The doṣa-herb framework is ancient. Modern science can test specific claims. Where does each work best? Where does each fall short?*” Each student writes 3 sentences.
5. **Linkage.** Module 13 (Indic Ecology) will revisit this through the *vasudhaiva kuṭumbakam* lens — herbs as part of the larger ecology.

Activity (15 min)

A scaled version of *Blind Smell-Test Identification* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Final Share and Assessment.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Formative Quiz — Herbs — Auṣadhi (Senior)

When: mid-module (after Day 5). **Duration:** 25 minutes. **Format:** mix of short answer, multiple choice, and reflection.

Questions

1. Name the three / four key terms introduced in this module so far. Give the IAST and one-line meaning of each.
2. State the central idea of Herbs — Auṣadhi in one sentence.
3. (Multiple choice) Which day in this module focused on ‘Neem — Bitter Healer’? (a) Day 1 (b) Day 2 (c) Day 3 (d) Day 4
4. True / False — and explain in one sentence: *Herbs — Auṣadhi is purely a religious concept and has no scientific framing.*
5. Connect Herbs — Auṣadhi to **one** other module in this curriculum. Which module, and how?
6. List **two** sources cited so far in this module.
7. (Short answer) Why does the module distinguish ‘auṣadhi’ from related concepts? Give one example.
8. (Short answer) In your own words, explain Day 3’s concept.
9. Identify **one** common misconception about Herbs — Auṣadhi and rebut it in one sentence.

10. Write **one** question you have about this module so far — something you want clarified before the summative.

How this is used

This quiz is **formative** — it's a check-in to see what's landing, not a grade. Teacher reviews answers to adjust pacing for Days 6–10. Students may swap and self-mark.

Summative Quiz — Herbs — Auṣadhi (Senior)

When: end of module (after Day 10). **Duration:** 45 minutes. **Total points:** 30.

Questions

Part A — Vocabulary (~5 points) A1. Define *auṣadhi* in your own words. A2. Define *tulasī* in your own words. A3. Define *nimba* in your own words. A4. Define *haridrā* in your own words. A5. Define *ārdra* in your own words.

Part B — Concepts (~10 points) B1. Explain the central idea of Herbs — Auṣadhi in your own words (3–4 sentences). B2. Describe the method or framework students used in Day 4 of this module. B3. Identify ONE source cited in this module and explain why it matters. B4. Compare Herbs — Auṣadhi with a concept from another module (your choice). What's similar, what's different? B5. Identify ONE common misconception about this module and rebut it in 2–3 sentences.

Part C — Application (~5 points) C1. Give an example of Herbs — Auṣadhi from your own daily life. C2. Describe one way this module connects to a current real-world issue (or scientific question).

Part D — Reflection (~5 points; ungraded for content) D1. What surprised you most in this module? D2. What's one question you still have?

Part E — Primary source (~5 points; Senior only) E1. Quote or paraphrase one passage from a primary source cited in this module. Explain it in your own words.

Marking guidance

- Vocabulary: give credit for correct meaning, even if exact wording differs.
- Concepts: look for evidence of understanding, not memorisation.
- Application: any reasonable example accepted.
- Reflection: ungraded for content; awarded for thoughtfulness.

Pass mark

A score of 18/30 indicates the student has met the module's learning objectives.

Homework

Homework — Herbs — Auṣadhi (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — What is Auṣadhi?** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — Tulsī — Holy Basil** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Neem — Bitter Healer** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — Turmeric — Golden Root** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Ginger — Warming Root** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Ajwain — Carom Seed** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Mint — Cooling Leaf** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Herbs and the Doṣas** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — Build the Class Deck** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Final Share and Assessment** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.

Writing Prompts

Writing Prompts — Herbs — Auṣadhi (Senior)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 300-word response.

1. In 3–4 sentences, explain Herbs — Auṣadhi to a friend who hasn't taken this module.
2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare Herbs — Auṣadhi with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.

7. What is one common misconception about Herbs — Auṣadhi? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?
9. Read one primary-source passage cited in this module. Translate or paraphrase it; comment on what it adds to your understanding.
10. Identify a contemporary debate about the IKS framework presented here (e.g., is Tirthaji’s mathematics genuinely Vedic? are doṣas a valid framework today?). Take a position with two pieces of evidence.

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Activity 1 — Blind Smell-Test Identification

Module: Herbs — Auṣadhi · **Band:** Senior · **Duration:** 30 min

Purpose

Reinforce the day-by-day concepts of Herbs — Auṣadhi through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

small samples of each of the six herbs, blindfolds (optional), notebook

What students do

Students close their eyes and try to identify each herb by smell. They record their guess + observation, then check.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (20 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Project Brief

Project Brief — Class Herb Identification Deck

Module: Herbs — Auśadhi · **Band:** Senior

What you will do

Each student creates one identification card for one of the six herbs: Sanskrit/botanical name, leaf drawing, smell description, three traditional uses, one active compound, one safety note. Cards combined into class library.

Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.
- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn't.
- **Final artefact** (a poster / model / digital deliverable).
- **Written reflection** (500 words (Senior)): what you learned, what surprised you, what you're still uncertain about.
- **Presentation** (5-minute share with the class).

Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.
- **Day 8:** document.
- **Day 9:** rehearse.
- **Day 10:** share day.

Sources

You must cite at least 3 sources. Use the form: *Author, Title, year*. (See `../sources.md` for examples.)

Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.

Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.

Rubric

Assessment Rubric — Herbs — Auṣadhi (Senior)

This rubric applies to the project (see `project-brief.md`) and may also inform the summative quiz.

Criteria

Accuracy of concepts — uses Sanskrit terms correctly, gets the central idea right

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Source discipline — cites sources, distinguishes traditional framing from empirical claim

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Application — connects the module to other modules and to real-world examples

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Presentation — clear, organised, evidence-supported

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Reflection — honest, thoughtful, identifies what was learned and what remains uncertain

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

How the teacher uses this rubric

1. After the project share, score each student on each criterion.
2. Convert to a band (4 = Exceeding, 3 = Meeting, etc.) for the report card.
3. Share at least one strength and one area for growth with each student.

Self-assessment (Senior students)

Senior students complete a parallel self-assessment using the same rubric before the teacher's assessment. The teacher and student then meet to compare.